PHASE 4 DATA SCIENCE PROJECT

Predicting Traffic Crash Causes in Chicago

Using Machine Learning to Help CDOT Allocate Safety Resources Smarter

430K+	3	6	5
Crash Records	Data Sources	Crash Categories	Models Tested

THE PROBLEM

Why Chicago Crashes Happen — and How to Predict Them

Business Challenge

The Chicago Department of Transportation (CDOT) currently responds reactively to traffic accidents. Without predictive insight into crash causes, safety resources are misallocated and high-risk patterns go undetected until lives are lost.

Goal: Build a model to predict the primary contributory cause of a crash — enabling proactive resource deployment.

Data Sources

Crashes.csv

Environmental factors, location, weather, road conditions

Vehicles.csv

Vehicle types, ages, mechanical defects

People.csv

Driver demographics, physical conditions

Over 360,000 rows labeled 'Unable to Determine' were filtered out — leaving 430K+ actionable records for modeling.

METHODOLOGY

Four-Phase Analytical Approach

01 Data Engineering

- Merged 3 datasets on CRASH_RECORD_ID
- Aggregated vehicle & people data to crash level
- Prevented Cartesian explosion via groupby aggregation

02 Target Engineering

- Mapped 30+ raw causes → 6 business categories
- Categories: Driver Error, Violation, Intoxication,
- Distraction, Environment, Vehicle Defect
- Dropped unclassifiable rows

03 Feature Pipeline

- Numeric: Median imputation + StandardScaler
- Categorical: UNKNOWN imputation + OneHotEncoder
- ColumnTransformer prevents data leakage

04 Modeling & Tuning

- 5 models across 2 iterations
- GridSearchCV with F1-Macro scoring
- Class weights to address severe imbalance

MODELING ITERATION JOURNEY

ITERATION 1 — White-Box Baselines

M1

Baseline Decision Tree

Accuracy: ~21%

Bias toward majority class

↗ *Upgrade to ensemble*

M2

Balanced Logistic Regression

Accuracy: Moderate

Linear & fully interpretable

ITERATION 2 — Ensemble & Tuning

M3

Random Forest (Iteration 1)

Accuracy: ~86%

Ignores minority classes

M4

Tuned Random Forest

Best for safety
★ RECOMMENDED

M5

Tuned Decision Tree

Accuracy: Moderate

Interpretable + minority recall

★ M4 Tuned Random Forest selected as primary production model — best minority-class recall for safety-critical decisions

MODEL PERFORMANCE

Accuracy, Precision, Recall & F1 Across All Models

Model	Accuracy	F1 Macro	F1 Weighted	Verdict
Baseline Decision Tree	**~21%**	Low	Low	❌ Unusable
Balanced Logistic Regression	**Moderate**	Moderate	Moderate	✅ Interpretable
Random Forest (Iteration 1)	**~86%**	High*	High*	⚠️ Ignores minority
Tuned Random Forest ★	**~40%**	**Balanced**	**Balanced**	✅ **Best Overall**
Tuned Decision Tree	**Moderate**	Moderate	Moderate	✅ Interpretable

** High F1 Weighted on majority class only — minority classes (INTOXICATION, VEHICLE_DEFECT) severely under-recalled.*

0.98 — DRIVER_ERROR Recall (RF Iter 1)

↑ Lifted — INTOXICATION Recall (Tuned RF)

F1-Macro — Tuning Metric (Grid Search)

KEY INSIGHTS

What the Data Reveals About Chicago Crashes

Driver Error Dominates

Driver Error accounts for the vast majority of actionable crashes, making it the single most impactful category for targeted CDOT campaigns and enforcement.

Environment & Weather Matter

Environmental features (road surface, weather, lighting) rank among the top feature importances — meaning infrastructure and seasonal campaigns have measurable ROI.

Minority Classes Are High-Risk

INTOXICATION and VEHICLE_DEFECT represent rare but catastrophically dangerous events. Missing these is costlier than dropping accuracy — justifying our balanced model choice.

Temporal Patterns Exist

Crash hour, day of week, and month all appear in the top features — peak risk windows can be identified, enabling time-targeted patrol and safety messaging.

Vehicle Age Is Predictive

Average vehicle age (computed from Vehicles dataset) drives differentiation in VEHICLE_DEFECT predictions — older fleets correlate with mechanical failure crashes.

Demographics Inform Context

Driver age (mean per crash) and physical condition provide secondary but meaningful signals, particularly for INTOXICATION and DISTRACTION classifications.

RECOMMENDATIONS

Strategic Next Steps for CDOT

1 IMMEDIATE

Deploy Tuned Random Forest

Use the tuned RF as the primary production model. Pair with the Tuned Decision Tree for interpretability briefings to non-technical stakeholders at safety board meetings.

2 SHORT-TERM

Address Minority Class Recall

Implement SMOTE (Synthetic Minority Oversampling) or advanced class-weight tuning to further lift INTOXICATION and VEHICLE_DEFECT recall — the two highest-cost misclassifications.

3 SHORT-TERM

Add SHAP Explainability Layer

Integrate SHAP values to generate per-prediction reason codes. This converts black-box outputs into actionable evidence for traffic officers and policy makers.

4 MEDIUM-TERM

Engineer Temporal Features

Parse CRASH_DATE to add rush-hour flags, weekend flags, and seasonal features. Temporal patterns in the data are strong signals currently left on the table.

5 MEDIUM-TERM

Benchmark XGBoost / LightGBM

Test gradient-boosted tree alternatives (XGBoost, LightGBM, CatBoost) against the tuned RF. Higher-capacity ensembles may close the accuracy/recall trade-off gap.

Primary Model: Tuned Random Forest • Interpretability Model: Tuned Decision Tree • Next Eval: Add SMOTE + SHAP

CONCLUSION

From Reactive Reporting to Predictive Safety

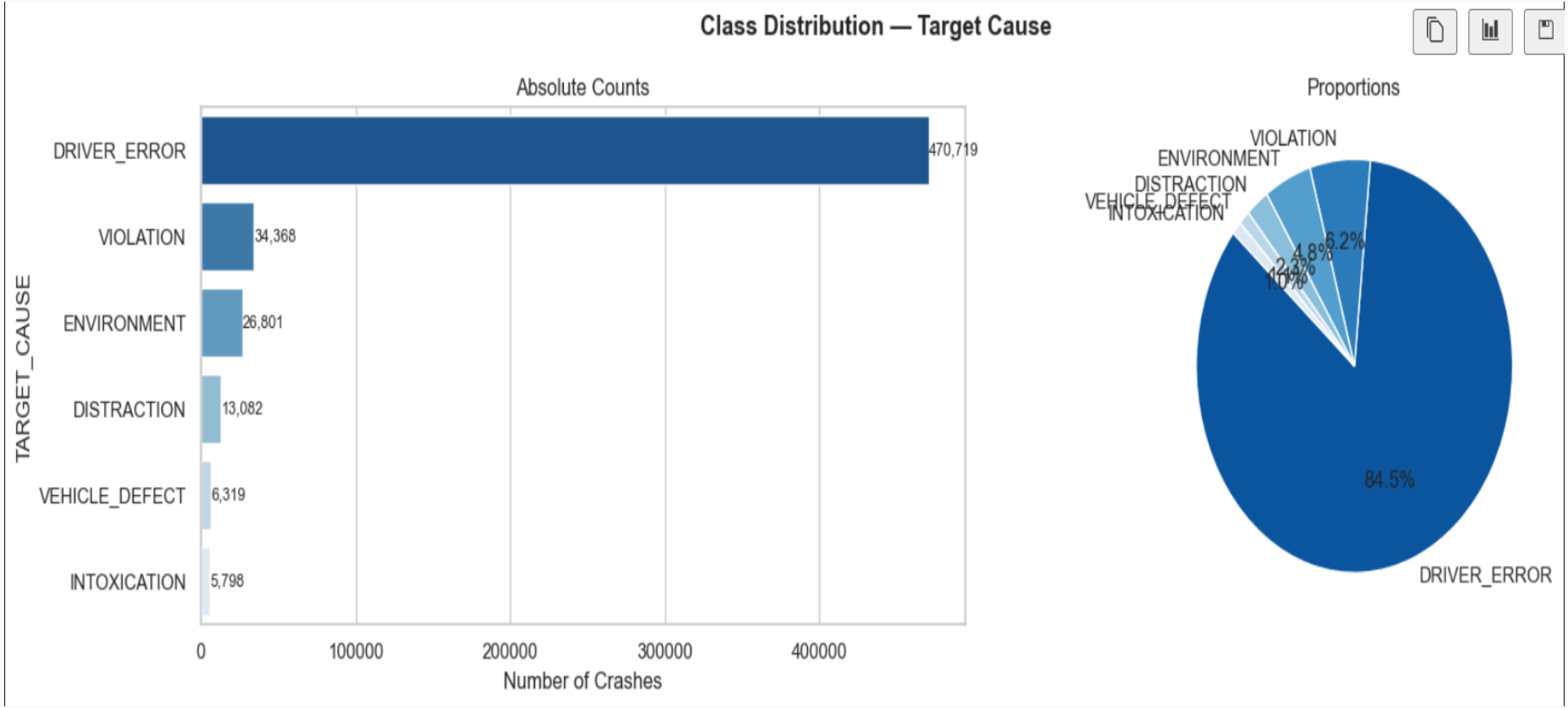
This project demonstrates that machine learning can transform Chicago's crash data into actionable intelligence. The Tuned Random Forest balances accuracy with safety-critical minority-class recall — enabling CDOT to move from after-the-fact reporting to proactive resource allocation.

By catching high-risk crash types before they become statistics, the model directly supports Chicago's mission to build safer roads for all residents.

Best Model	Key Driver	Priority Fix
Tuned Random Forest	**Driver Error (70%+)**	**SMOTE + SHAP**

VISUALISATIONS

Exploratory Data Analysis



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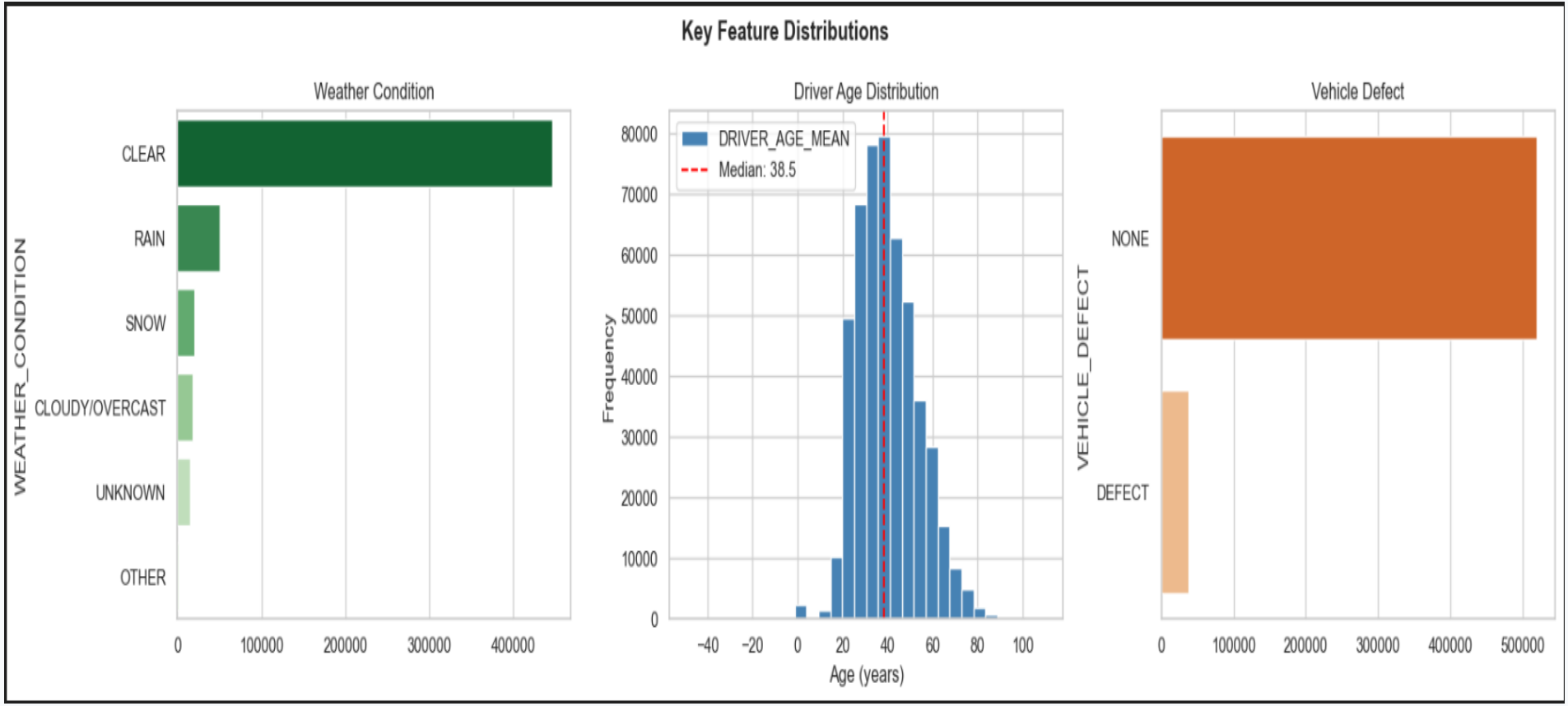
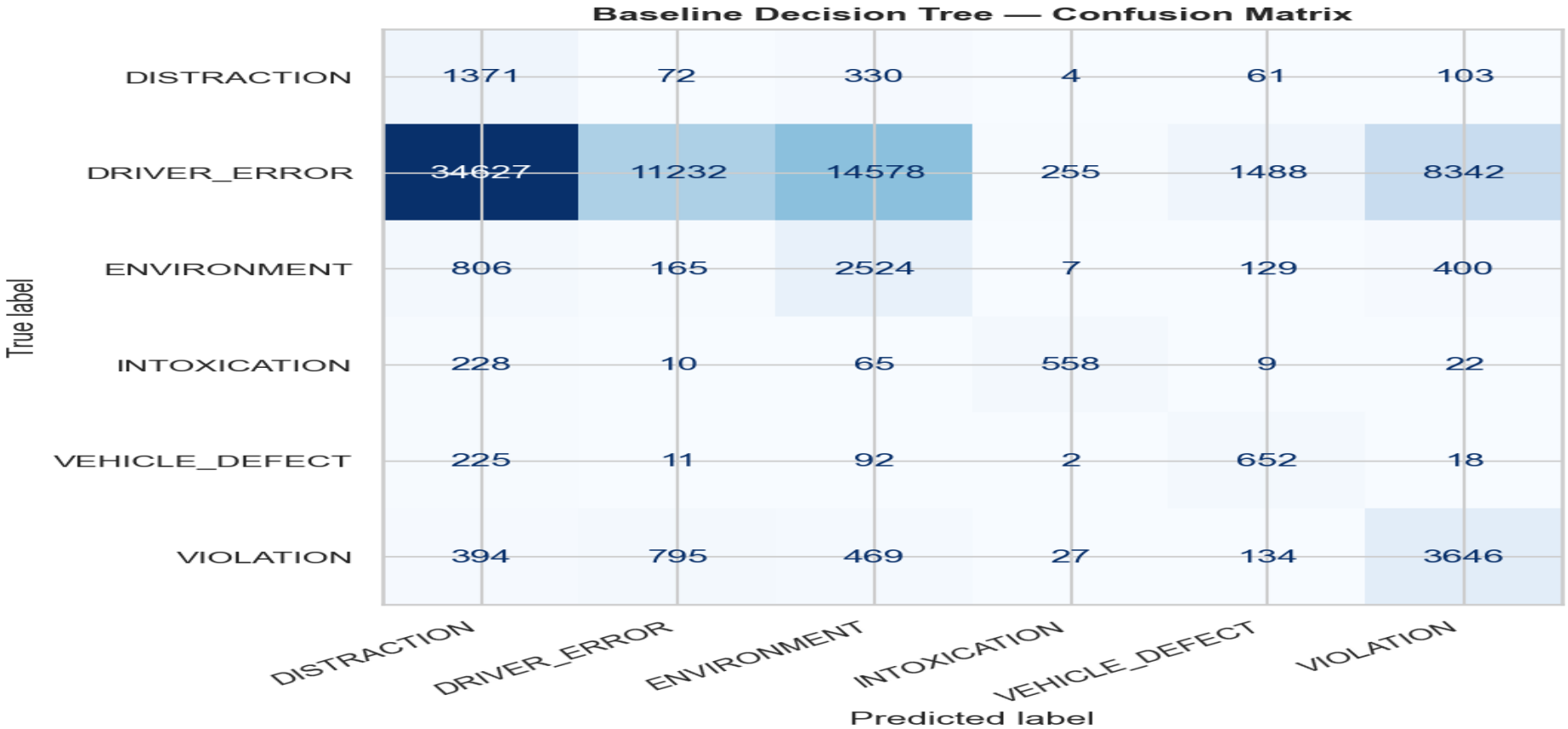


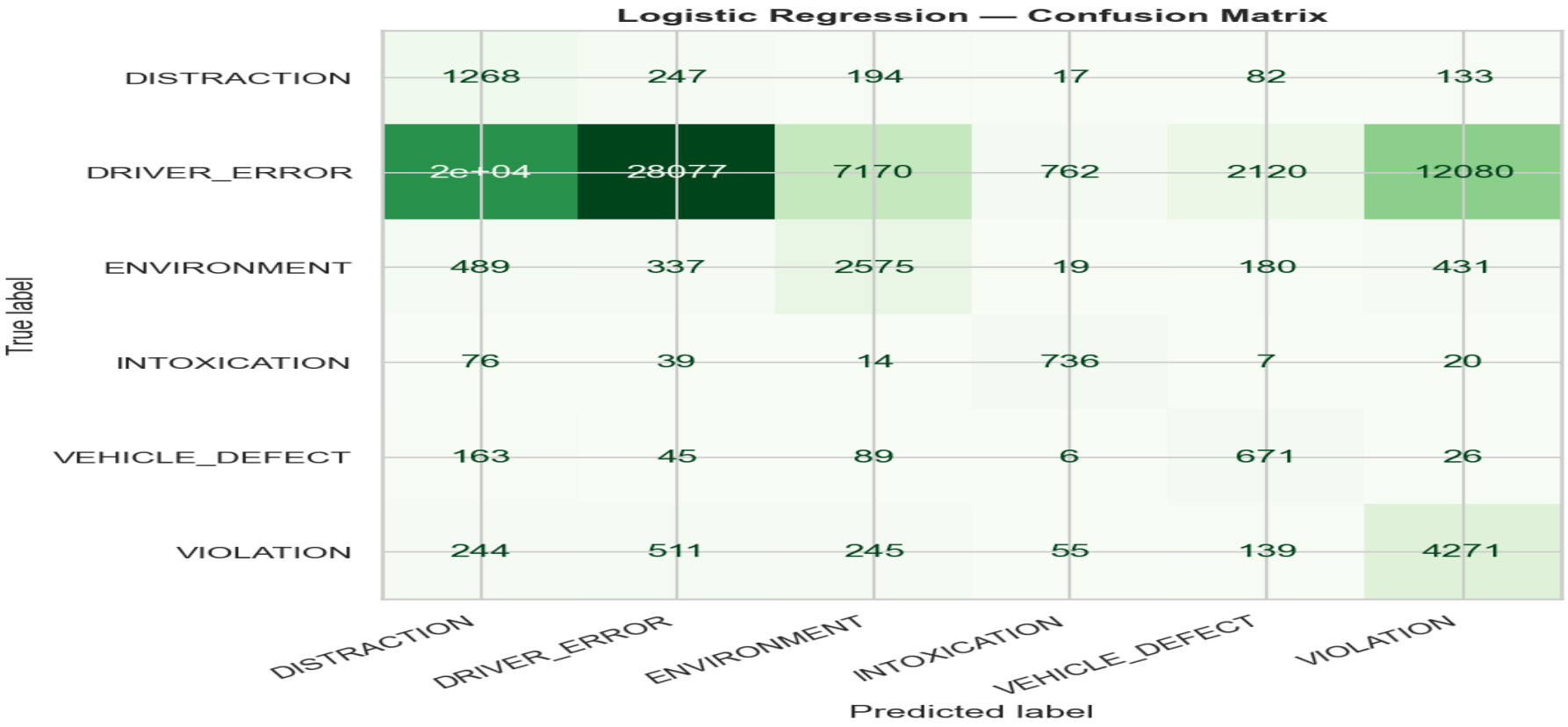
VISUALISATIONS

Model 1 — Baseline Decision Tree



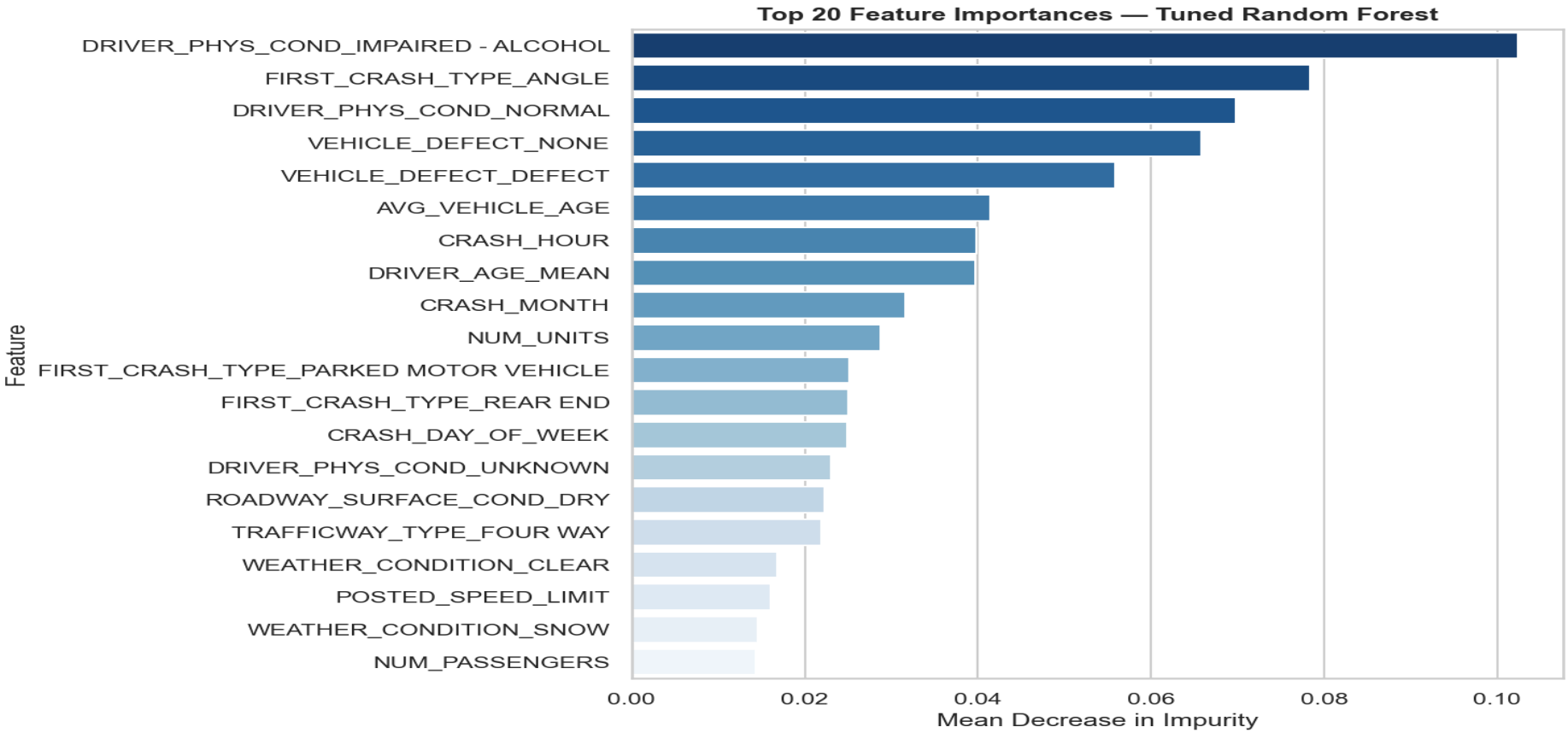
VISUALISATIONS

Model 2 — Balanced Logistic Regression



VISUALISATIONS

Feature Importance — Tuned Random Forest



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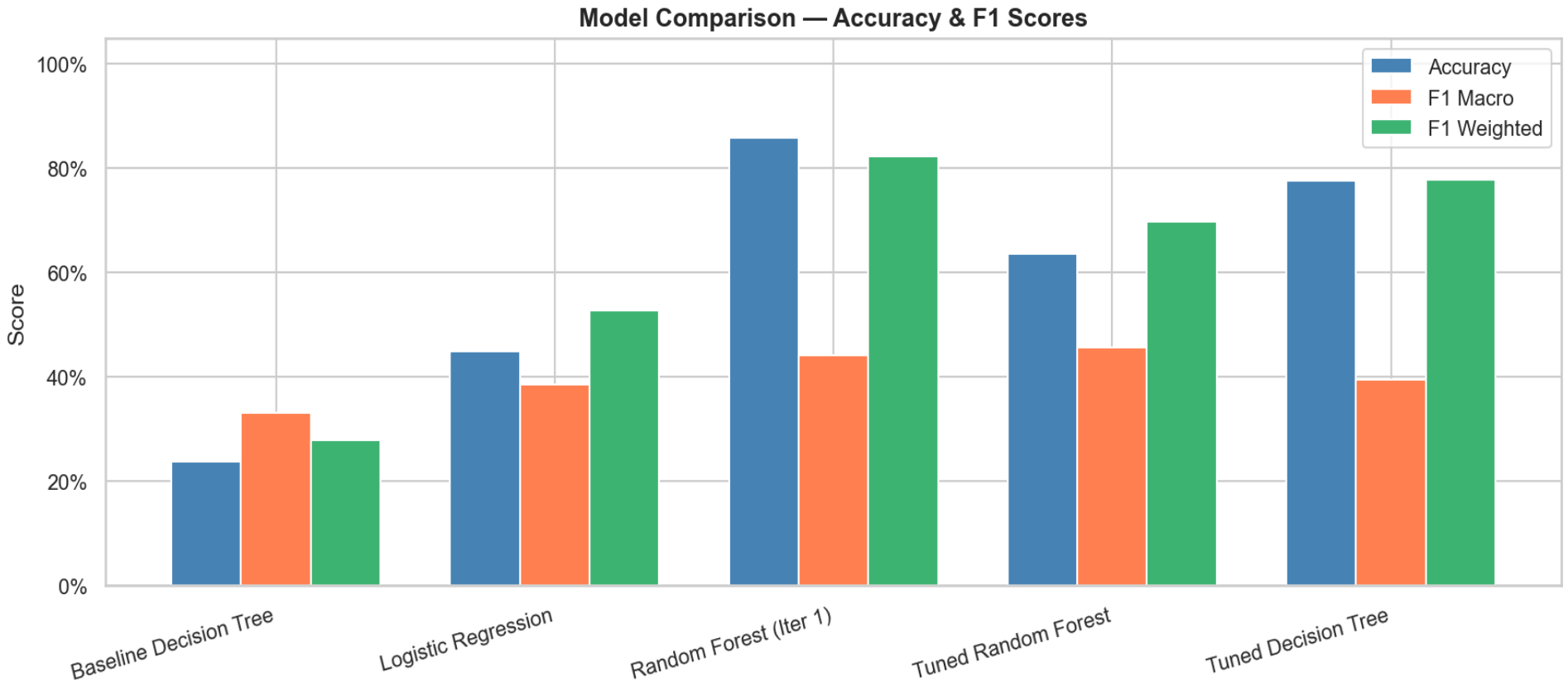
Model Comparison Summary